

CSE260

Assignment 03 (Total Marks: 25)

This assignment must be hand-written. Show ALL steps in ALL questions.

Make sure that your circuit is efficient, meaning you should use the lowest number of components. You may use external gates if required.

Answer any 5 questions

Question 1 (5 Marks)

Build a circuit that implements the 1's complement number system (3 bit) using encoder(s) and decoder(s).

Question 2: (5 Marks)

Implement the following boolean function using **a)** single 16:1 mux **b)** single 8:1 mux

$$F(A,B,C,D) = \sum(0,1,2,7,8,10,11,13, 15)$$

Use external gates if required.

Question 3: (5 Marks)

Implement the following boolean function using **a)** 4x16 decoder(s) only **b)** 2x4 decoder(s) only

$$F(A,B,C,D,E) = \sum(0,1,2,7,8,10,11,13, 15,18,21,24,25)$$

Use external gates if required.

Question 4: (5 Marks)

Design a 13 person voter counting system using full and parallel adders.

Question 5: (5 Marks)

Consider A is a 4 bit number. Design A-3 (A minus 3) using a 4 bit parallel adder. Use external gates if required.

Question 6 (5 Marks)

Implement the following using only one 4x1 mux $F(A,B,C,D) = \sum (0,1,2,8,10,11,14,15)$

Question 7 (5 Marks)

You want to build a calculator that can add or subtract two 4-bit numbers: X and Y. Now, if both of the numbers are greater than or equal to 8, then the calculator will do the operation X+Y (addition). Otherwise, the calculator will do the operation X-Y(subtraction). **Design** the circuit using a 4-bit parallel adder cum subtractor.

